INCH-POUND

MIL-DTL-17587/3(SH)
28 June 2007

DETAIL SPECIFICATION SHEET

CIRCUIT BREAKERS, ACB, LOW VOLTAGE, ELECTRIC POWER, AIR, REMOVABLE CONSTRUCTION ACB-2000HR, ACB-2000LRC, ACB-2000RC, ACB-2002HRC, ACB-2004HRC, ACB-2010HRC, ACB-2012, AND ACB-2020

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17587.

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REQUIREMENTS:

TABLE I. Ratings/features of circuit breakers.

Circuit breaker type	ACB-2000HR (PIN 1)	ACB-2000HR (PIN 2)	ACB-2000LRC	ACB-2000RC	ACB-2002HRC	ACB-2004HRC	ACB-2010HRC	ACB-2012	ACB-2020
Rated continuous current (amperes)	(2000)	2000	2000	2000	2000	2000	2000	2000	2000
Rated maximum voltage (AC volts) /1	500	500	500	500	500	500	500	500	500
Frequency (hertz)	60	60	60	60	60	60	60	60	60
Maximum interrupting rating current (symmetrical amperes)	85,000	85,000	85,000	50,000	100,000	100,000	85,000	85,000	85,000
Rated short-time current /2 (amperes)	85,000	85,000	70,000	50,000	100,000	100,000	85,000	85,000	85,000
Rated short-time current duration (seconds)	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Operating mechanism - manual	STD	STD	STD	STD	STD	STD	STD	STD	STD
Operating mechanism - electric	STD	STD	STD	STD	STD	STD	STD	STD	STD
Maximum weight (lbs) - circuit breaker (removable component)	505	318	423	423	449	450	205	202	200
Maximum weight (lbs) - cradle (stationary component)	209	180	209	209	182	185	278	130	120
Number of poles	3	3	3	3	3	3	3	3	3
Number of overcurrent coils or solid-state sensors	2/3	3	2/3	2/3	2/3	3	3	3	3 – 1 per pole

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TABLE I. Ratings/features of circuit breakers - Continued.

Circuit breaker type	ACB-2000HR (PIN 1)	ACB-2000HR (PIN 2)	ACB-2000LRC	ACB-2000RC	ACB-2002HRC	ACB-2004HRC	ACB-2010HRC	ACB-2012	ACB-2020
Attachments									
Electric closing mechanism	STD	STD	STD	OPT	STD	STD	STD	OPT	STD
Auxiliary switches (number of contacts)	8	10	8	8	12	12	8	8	8
Undervoltage lockout device	Optional for all circuit breaker types.							N/A	N/A
Mechanical position indicator	Optional for all circuit breaker types.							STD	STD
Secondary disconnecting devices	Required for all circuit breaker types, unless otherwise specified.								
Cell switch	Optional for all circuit breaker types.							N/A	STD
Indicator light	Provisions for mounting required for all circuit breaker types; furnished only when specified.							OPT	OPT
Hold-closed mechanism	OPT	OPT	OPT	OPT	OPT	N/A	N/A	N/A	N/A
Undervoltage release device	Optional for all circuit breaker types.								N/A
Shunt trip device	Required for electrically-operated breakers.							OPT	STD
Operations counter	OPT	OPT	OPT	OPT	OPT	OPT	OPT	OPT	STD
Trip on power up (TOPU) device	N/A	N/A	N/A	N/A	OPT	OPT	OPT	N/A	N/A
Rack-in/rack-out position indicator	STD	STD	STD	STD	STD	STD	STD	STD	STD

NOTES:
/1 Voltage ratings apply to main current-carrying parts. Operating attachments shall be of voltages specified in 6.2 of MIL-DTL-17587.
/2 Ampere values apply to circuit breakers with overcurrent coils or sensors of maximum current rating within the circuit breaker type.

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TABLE II. Available continuous current ratings and pickup current settings for overcurrent coils and sensors.

Type	Coil or sensor rating /1 (amperes)	Long-time pickup setting (amperes)			Short-time pickup setting (amperes)					Maximum instantaneous trip /2 (amperes)
ACB-2000HR (PIN 1)	2000	3000	4000	5000	4000	6000	8000	10000	---	80000
ACB-2000HR (PIN 2)	320	480	640	800	640	800	960	1280	1600	80000
	400	600	800	1000	800	1000	1200	1600	2000	80000
	480	720	960	1200	960	1200	1440	1920	2400	80000
	560	840	1120	1400	1120	1400	1680	2240	2800	80000
	640	960	1280	1600	1280	1600	1920	2560	3200	80000
	800	1200	1600	2000	1600	2000	2400	3200	4000	80000
	1000	1500	2000	2500	2000	2500	3000	4000	5000	80000
	1200	1800	2400	3000	2400	3000	3600	4800	6000	80000
	1400	2100	2800	3500	2800	3500	4200	5600	7000	80000
	1600	2400	3200	4000	3200	4000	4800	6400	8000	80000
	2000	3000	4000	5000	4000	5000	6000	8000	10000	80000
ACB-2000LRC	2000	3000	4000	5000	4000	6000	8000	10000	---	68000

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TABLE II. Available continuous current ratings and pickup current settings for overcurrent coils and sensors - Continued.

Type	Coil or sensor rating /1 (amperes)	Long-time pickup setting (amperes)			Short-time pickup setting (amperes)					Maximum instantaneous trip /2 (amperes)
ACB-2000RC	320	500	640	840	---	640	960	1280	1600	16000
	400	640	840	960	---	800	1280	1800	2100	20000
	480	720	960	1120	---	960	1600	2100	2800	24000
	560	840	1120	1280	---	1120	1800	2400	3200	28000
	640	960	1280	1600	---	1280	2100	2800	3600	32000
	800	1280	1600	2100	1200	1600	2400	3200	4000	36000
	1000	1600	2100	2400	---	2100	3200	4000	4800	40000
	1200	1800	2400	2800	1800	2400	3600	4800	6000	40000
	1400	2100	2800	3600	---	2800	3600	4800	6000	40000
	1600	2400	3200	4000	2400	3200	4800	6000	8000	40000

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TABLE II. Available continuous current ratings and pickup current settings for overcurrent coils and sensors – Continued.

Type	Coil or sensor rating /1 (amperes)	Long-time pickup setting (amperes)			Short-time pickup setting (amperes)					Maximum instantaneous trip /2 (amperes)
ACB-2002HRC, ACB-2004HRC	200	300	400	500	400	600	800	1000	---	4000
	300	450	600	750	600	900	1200	1500	---	6000
	500	750	1000	1250	1000	1500	2000	2500	---	10000
	800	1200	1600	2000	1600	2400	3200	4000	---	16000
	1200	1800	2400	3000	2400	3600	4800	6000	---	24000
	1600	2400	3200	4000	3200	4800	6400	8000	---	32000
	2000	3000	4000	5000	4000	6000	8000	10,000	---	40000
ACB-2010HRC	800	1200	1600	2000	1600	2000	2400	3200	4000	40000
	1000	1500	2000	2500	2000	2500	3000	4000	5000	40000
	1200	1800	2400	3000	2400	3000	3600	4800	6000	40000
	1400	2100	2800	3500	2800	3500	4200	5600	7000	40000
	1600	2400	3200	4000	3200	4000	4800	6400	8000	40000
	2000	3000	4000	5000	4000	5000	6000	8000	10000	40000

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TABLE II. Available continuous current ratings and pickup current settings for overcurrent coils and sensors – Continued.

Type	Coil or sensor rating /1 (amperes)	Long-time pickup setting (amperes)	Short-time pickup setting (amperes)	Maximum instantaneous trip /2 (amperes)
ACB-2012	2000 /3	See Figure 12	See Figure 12	See Figure 12
ACB-2020	1250 /4	See Figure 16	See Figure 16	See Figure 16
	1600 /4			
	2000 /4			
	2500 /4			
	3200 /4			

NOTES:

/1 Continuous current rating unless otherwise noted.

/2 The instantaneous trip device shall be set at the place of manufacture from five times the coil or sensor rating to the maximum pickup setting specified for mechanical coils and 20 times sensor rating for solid-state sensors with a default at maximum listed rating.

/3 The only available sensor rating is 2000 amps but the continuous current rating is selectable from 500 to 2000 amps in 100-amp increments.

/4 The coil or sensor ratings shown correspond to the following continuous current ratings:

Coil or sensor rating	Continuous current rating
1250	800
1600	1000
2000	1200
2500	1600
3200	2000

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TABLE III. Electrical endurance and temperature rise test requirements.

Electrical endurance test				
Type	**Total number of close-open cycles**	**Amperes on circuit breaker close**	**Amperes on circuit breaker open**	**Maximum AC power factor (lagging) X/R**
ACB-2000HR (PIN 1)	1000	8000	2000	Circuit breaker close: 0.25 Circuit breaker open: 0.80
ACB-2000HR (PIN 2)	1000	8000	2000	
ACB-2000LRC	10,000	8000	2000	
ACB-2000RC	1000	2000	2000	
ACB-2002HRC	10,000	8000	2000	
ACB-2004HRC	10,000	8000	2000	
ACB-2010HRC	1000	8000	2000	
ACB-2012	5000	2000	2000	0.80
ACB-2020	1000	2000	2000	0.80

Temperature rise test					
Type	**Continuous current rating**	**Cable quantity per terminal /1**	**Cable size**	**Bus bar quantity per phase**	**Bus bar size (inches) /2**
ACB-2000HR (PIN 1)	2,000	5	400 MCM	2	4.0 x ¼
ACB-2000HR (PIN 2)					
ACB-2000LRC					
ACB-2000RC					
ACB-2002HRC					
ACB-2004HRC					
ACB-2010HRC					
ACB-2012					
ACB-2020					

NOTES:

/1 Information regarding cable is contained in MIL-DTL-915. If 3-phase cable is used for this test, the cable jacket shall be stripped for a minimum of 2 feet.

/2 Information regarding bus bars is contained in MIL-DTL-16036.

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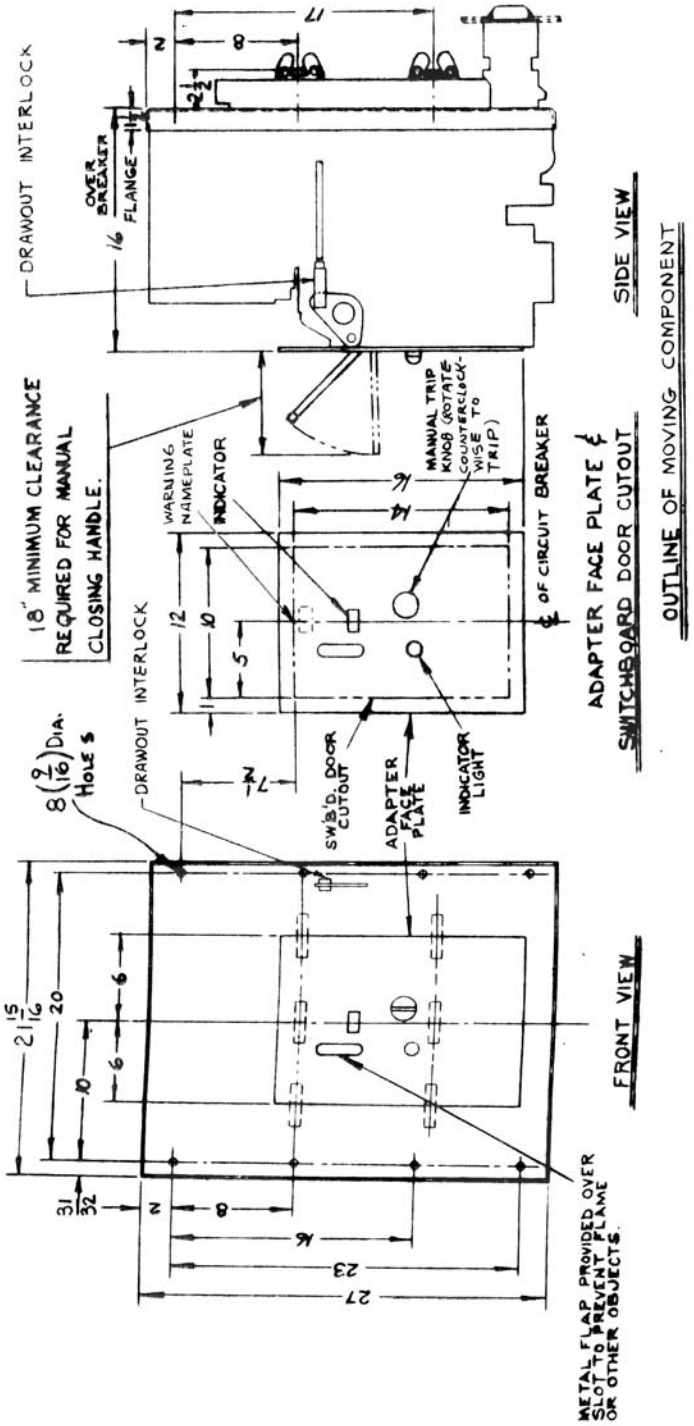


FIGURE 1. Outline and mounting dimensions for circuit breaker types ACB-2000RC, ACB-2000HR (PIN 1), and ACB-2000LRC.

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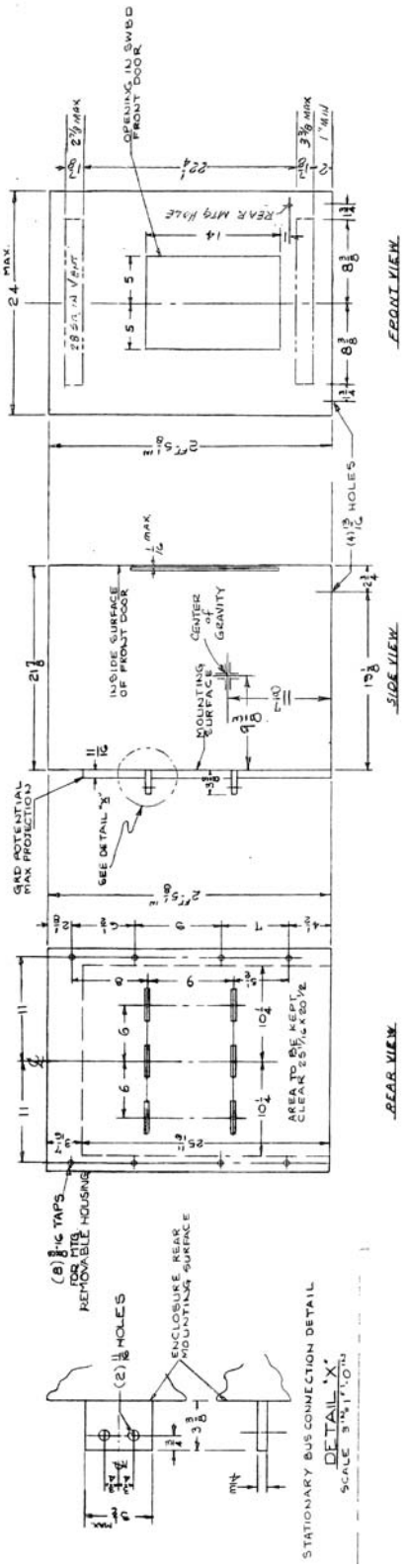


FIGURE 2. Outline and mounting dimensions for circuit breaker types ACB-2000RC, ACB-2000HR (PIN 1), and ACB-2000LRC.

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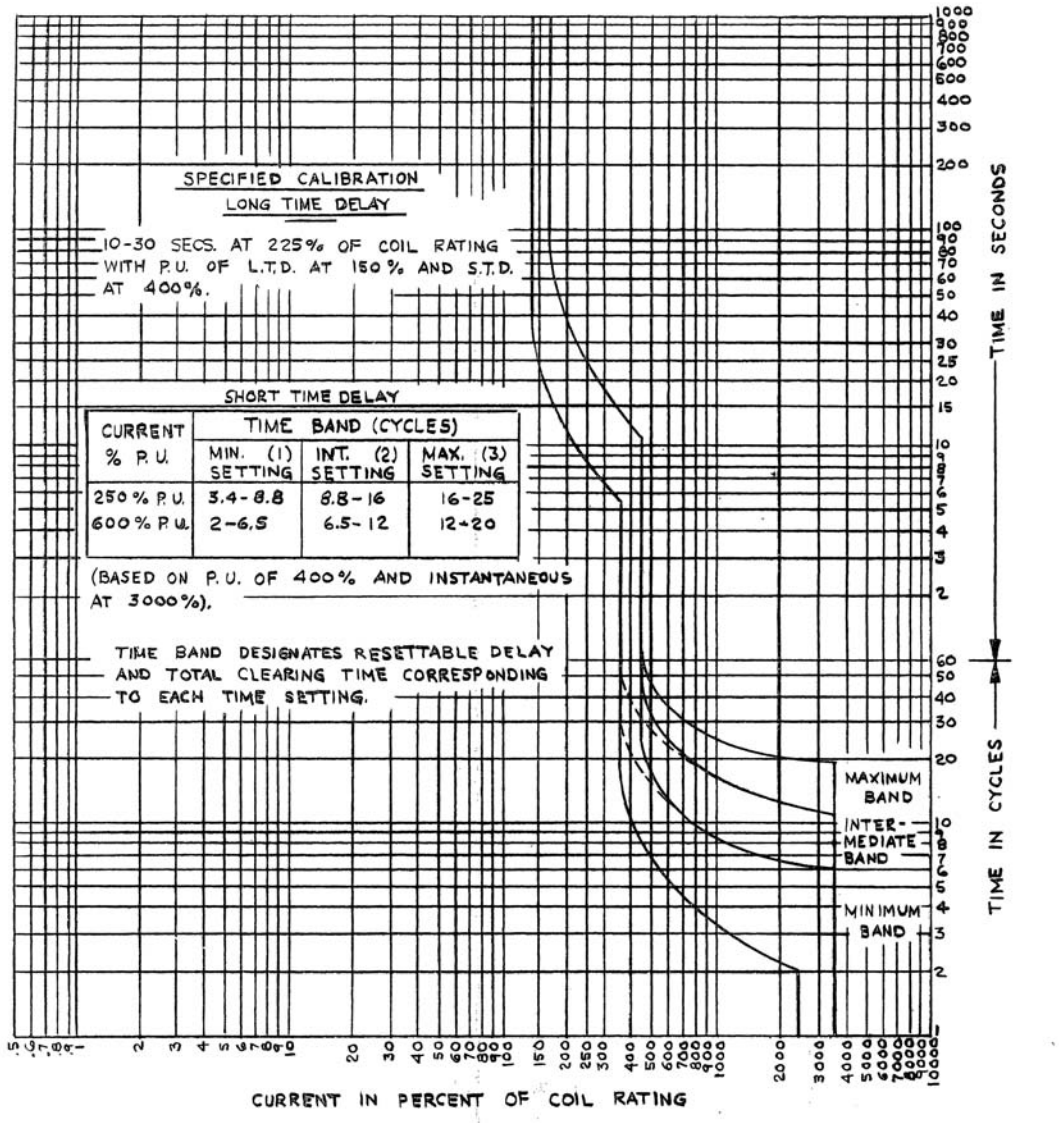


FIGURE 3. Time-current characteristic curve for circuit breaker types ACB-2000RC, ACB-2000HR (PIN 1), and ACB-2000LRC.

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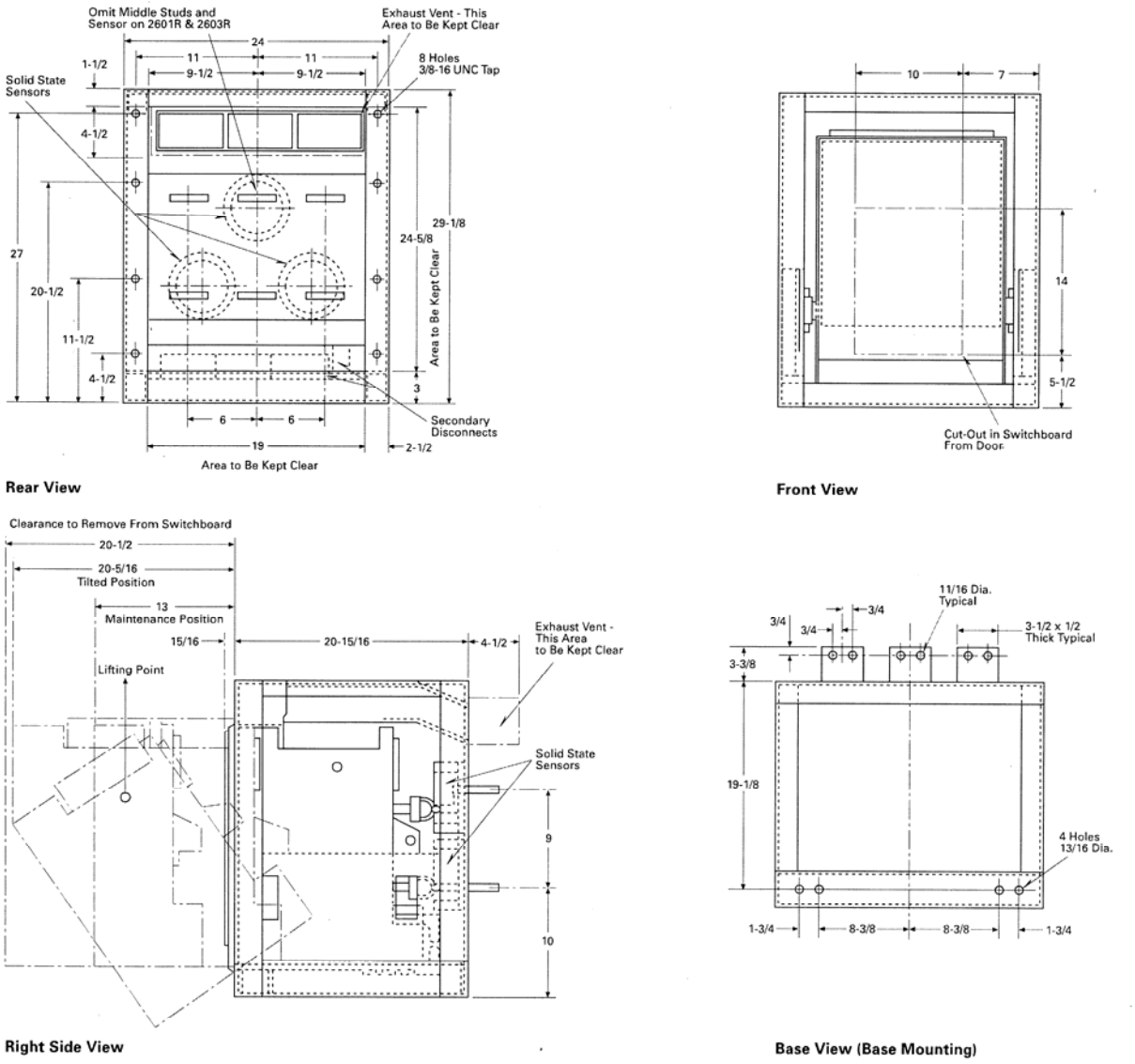


FIGURE 4. Outline and mounting dimensions for circuit breaker type ACB-2000HR (PIN 2).

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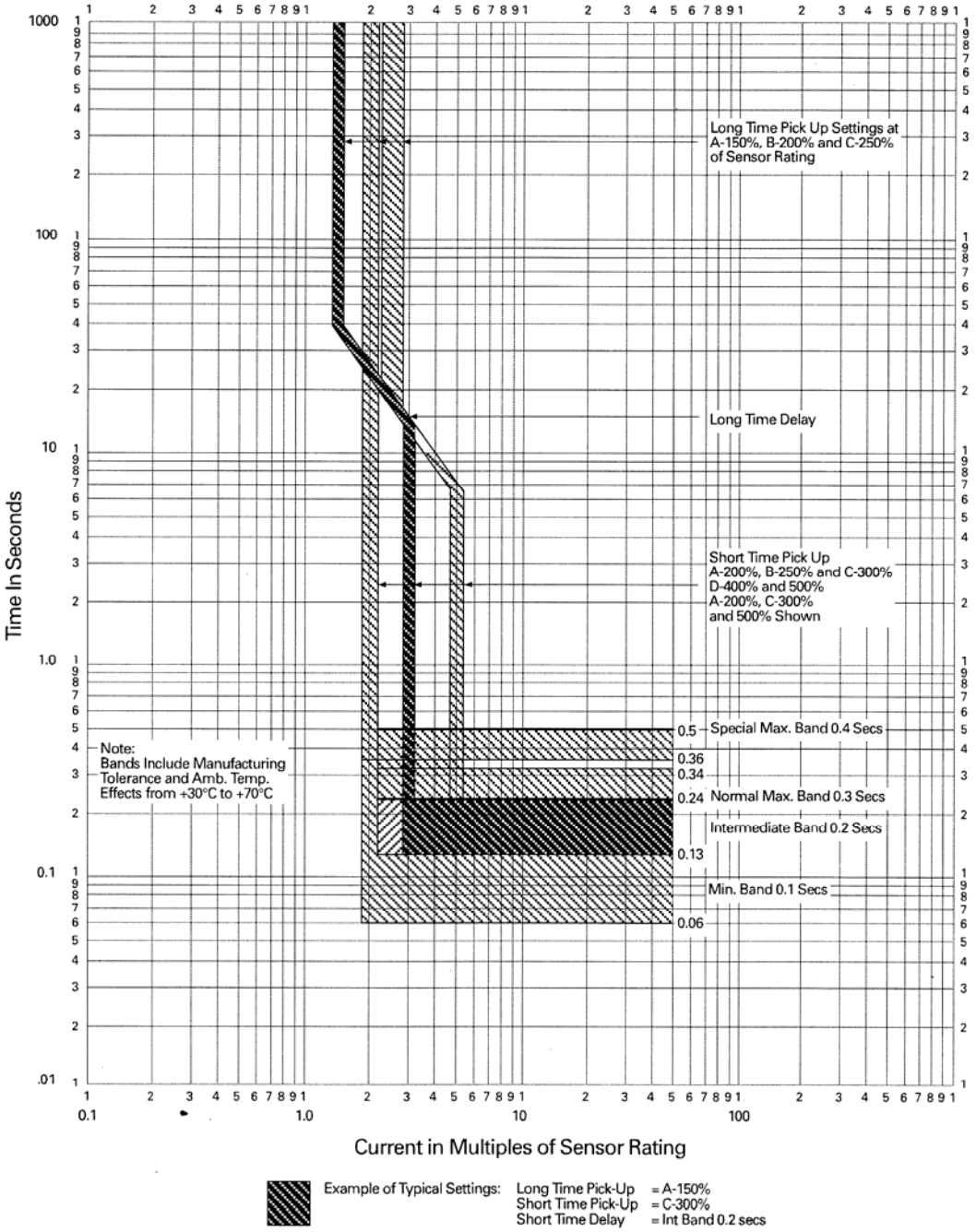


FIGURE 5. Time-current characteristic curve for circuit breaker type ACB-2000HR (PIN 2).

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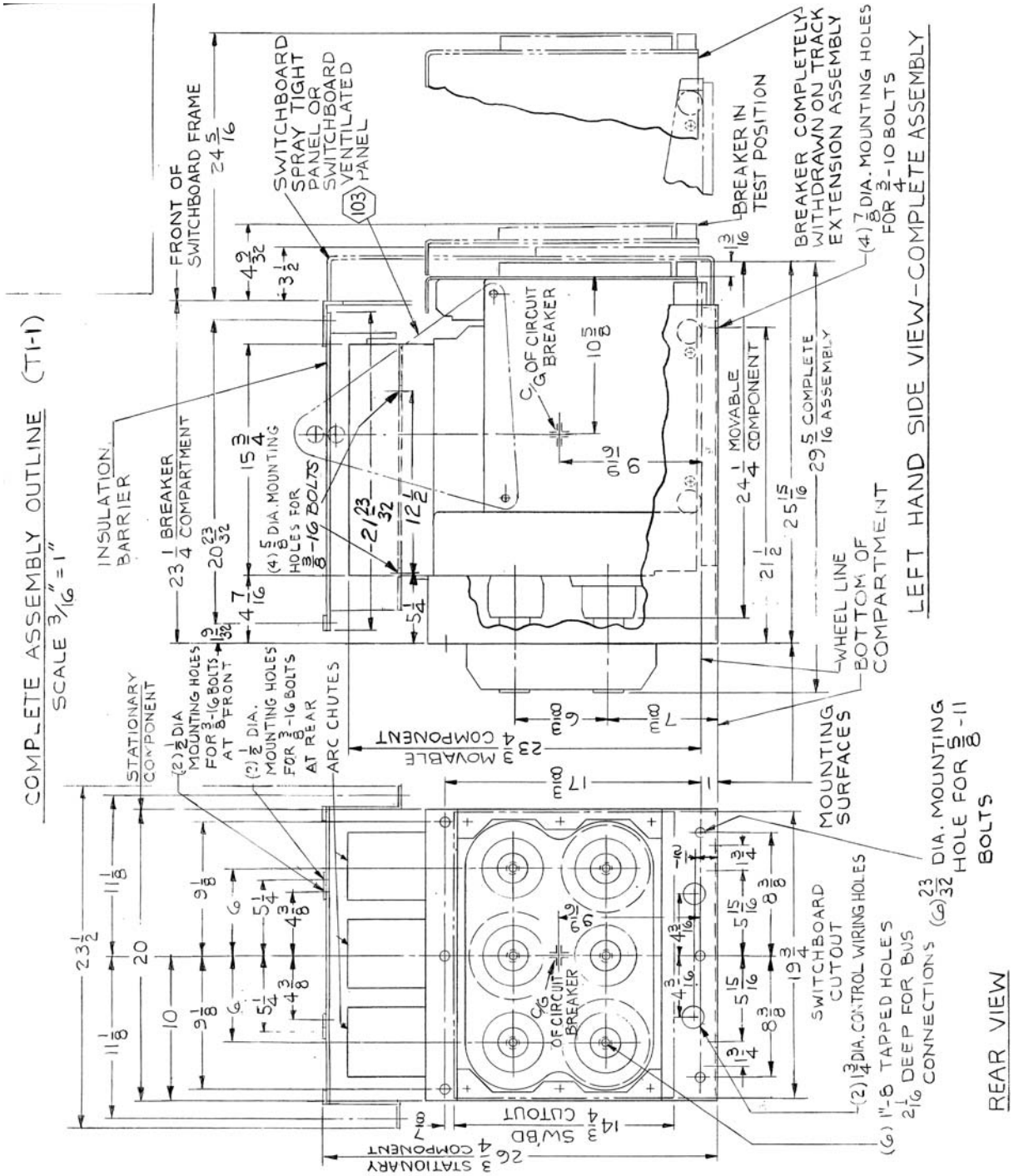


FIGURE 6. Outline and mounting dimensions for circuit breaker types ACB-2002HRC and ACB-2004HRC.

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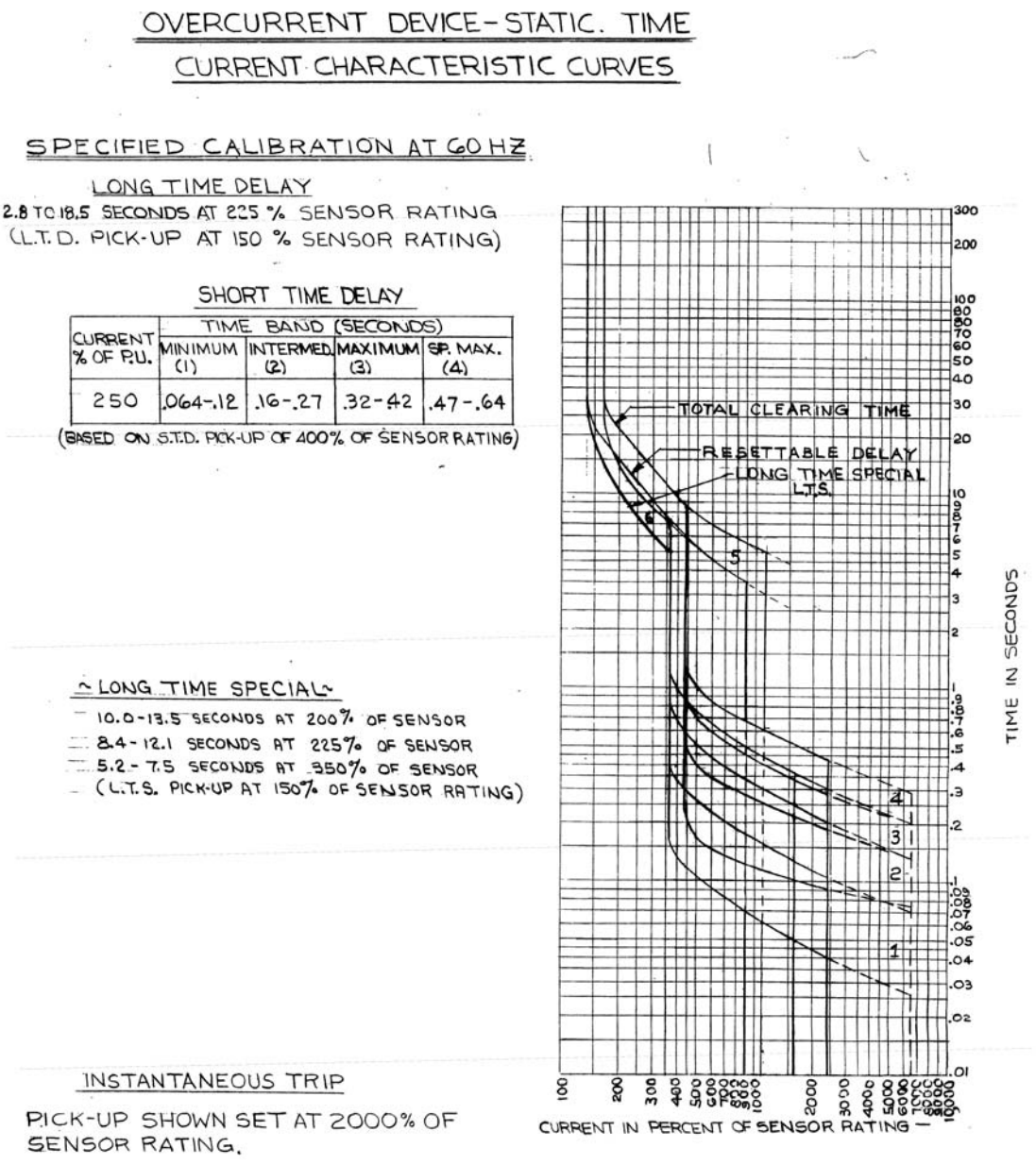


FIGURE 7. Time-current characteristic curve for circuit breaker types ACB-2002HRC and ACB-2004HRC.

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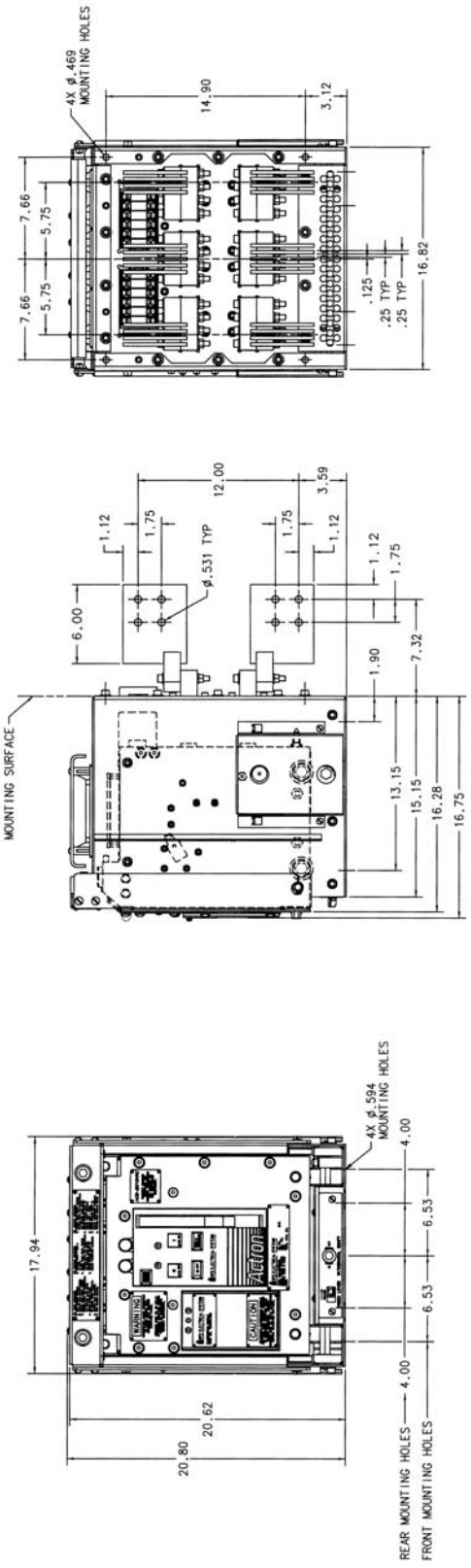


FIGURE 8. Outline and mounting dimensions for circuit breaker type ACB-2010HRC.

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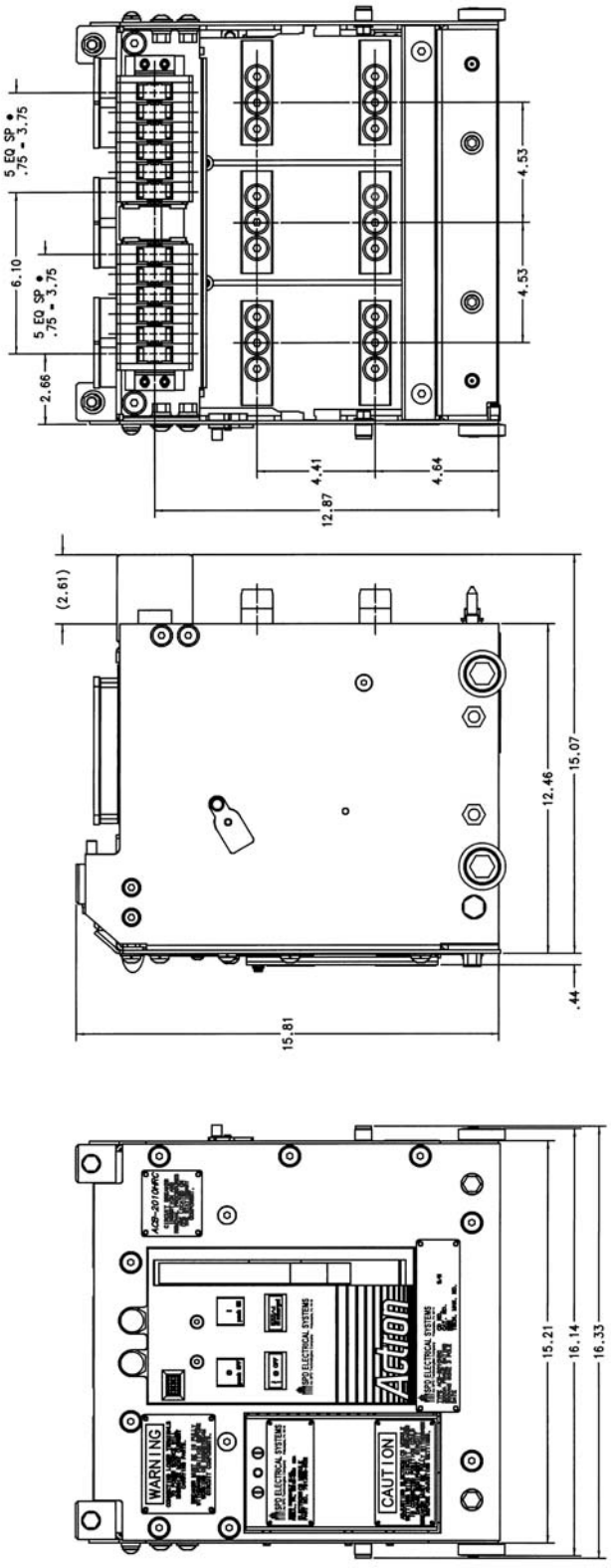


FIGURE 9. Outline and mounting dimensions for circuit breaker type ACB-2010HRC.

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ADJUSTABLE RATINGS AND CONTINUOUS CURRENT SETTINGS (CCS)					
800A	1000A	1200A	1400A	1600A	2000A

ADJUSTABLE LONG-TIME DELAY PICKUP SETTINGS (IN MULTIPLES OF CONTINUOUS CURRENT SETTINGS)		
1.5X	2.0X	2.5X

ADJUSTABLE SHORT-TIME DELAY PICKUP SETTINGS (IN MULTIPLES OF CONTINUOUS CURRENT SETTINGS)				
2.0X	2.5X	3.0X	4.0X	5.0X

ADJUSTABLE INSTANTANEOUS PICKUP SETTINGS (IN MULTIPLES OF 2000 AMPERE FRAME RATING)						
3X	5X	6X	8X	10X	12X	15X	18X	20X
6000	10000	12000	16000	20000	24000	30000	36000	40000

ADJUSTABLE SHORT-TIME DELAY SETTINGS	
SHORT-TIME DELAY BAND	DELAY (SECONDS)
1	.065-.130
2	.130-.240
3	.240-.340
4	.360-.500

PER MIL-C-17587

LONG-TIME AND SHORT-TIME PICKUP CURRENTS ARE ACCURATE TO ±10%. INSTANTANEOUS PICKUP CURRENTS ARE ACCURATE TO:

±10% FOR PICKUPS ≤ 10X FRAME RATING
±20% FOR PICKUPS > 10X FRAME RATING

LONG-TIME AND SHORT-TIME CALIBRATION ARE GUARANTEED OVER TEMPERATURE RANGE DEFINED BY SWITCHGEAR AMBIENT OF 10°C TO 50°C.

INSTANTANEOUS TRIP POINTS ARE GUARANTEED AT 25°C ONLY. CONSULT FACTORY FOR ACCURACY AT HIGHER TEMPERATURE.

TRIP TIMES ARE FROM INITIATION OF SYMMETRICAL SINUSOIDAL CURRENT TO EXTINGUISHING OF ARC AT RATED VOLTAGE.

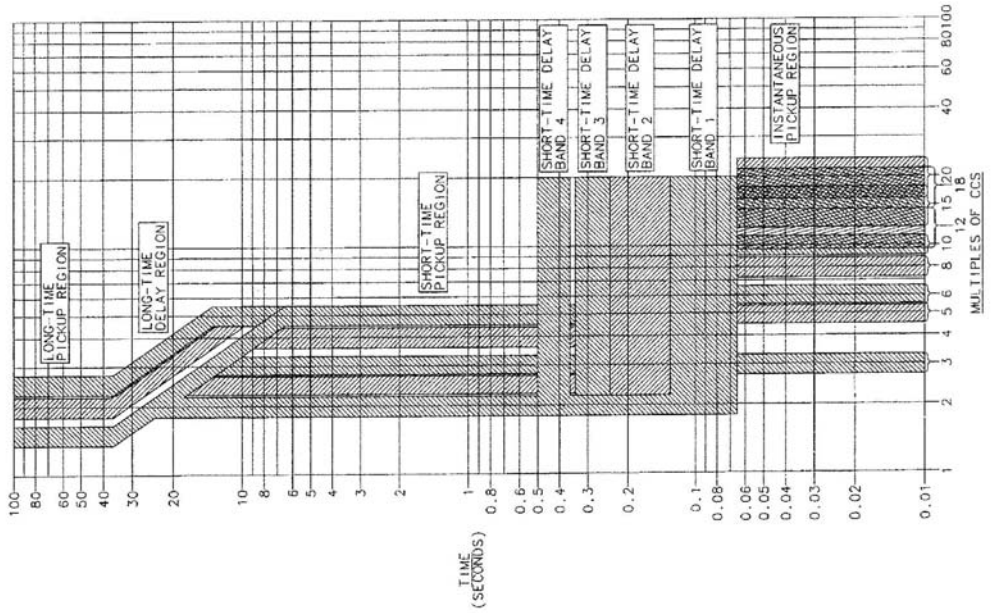


FIGURE 10. Time-current characteristic curve for circuit breaker type ACB-2010HRC.

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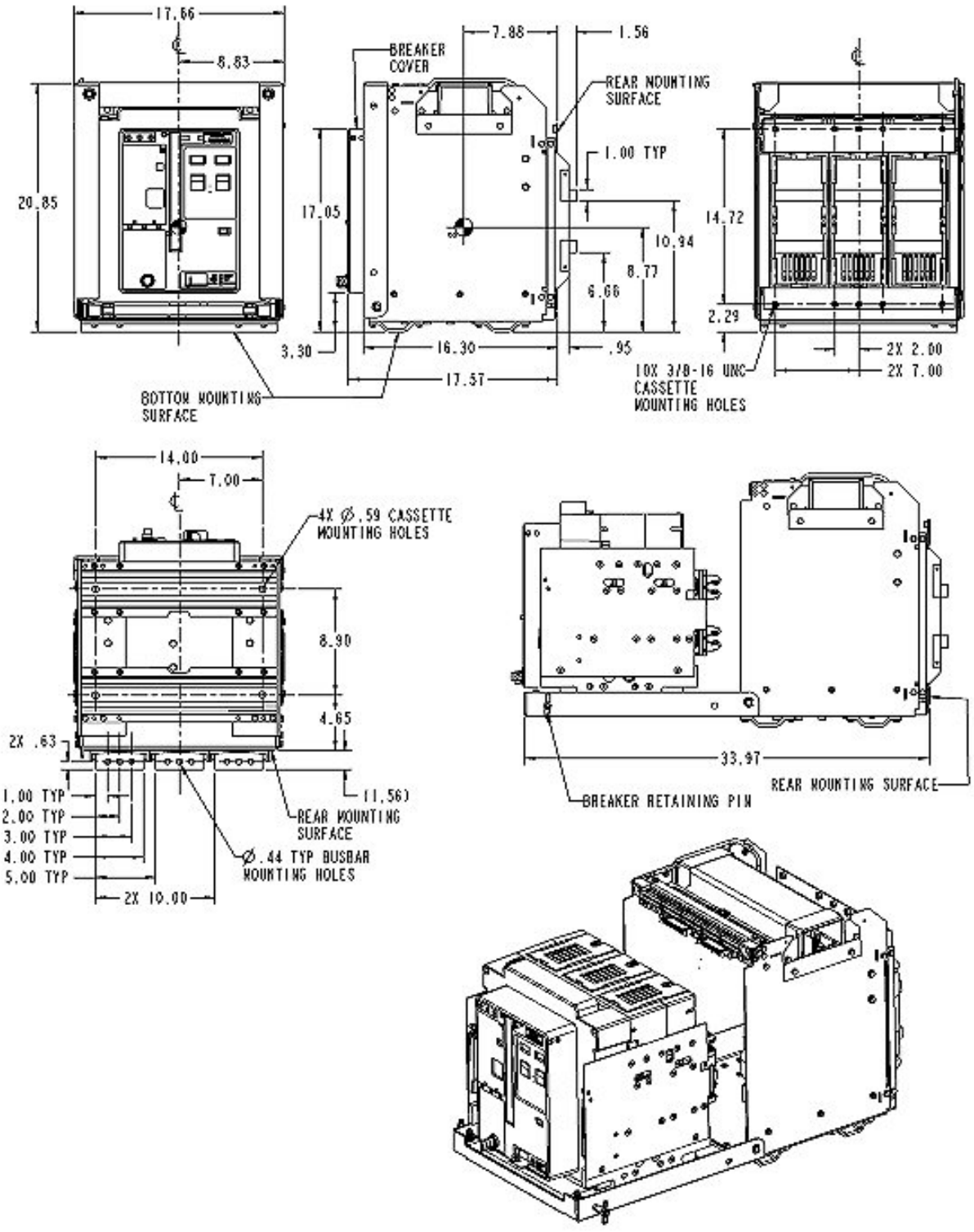


FIGURE 11. Outline and mounting dimensions for circuit breaker type ACB-2012.

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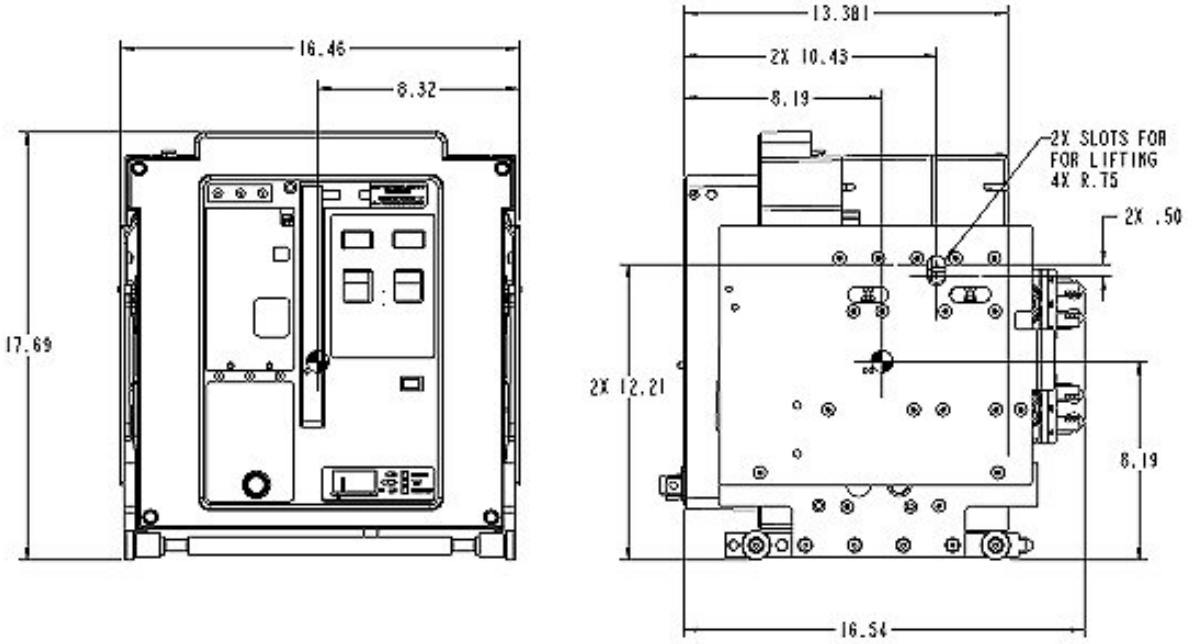


FIGURE 11. Outline and mounting dimensions for circuit breaker type ACB-2012 - Continued.

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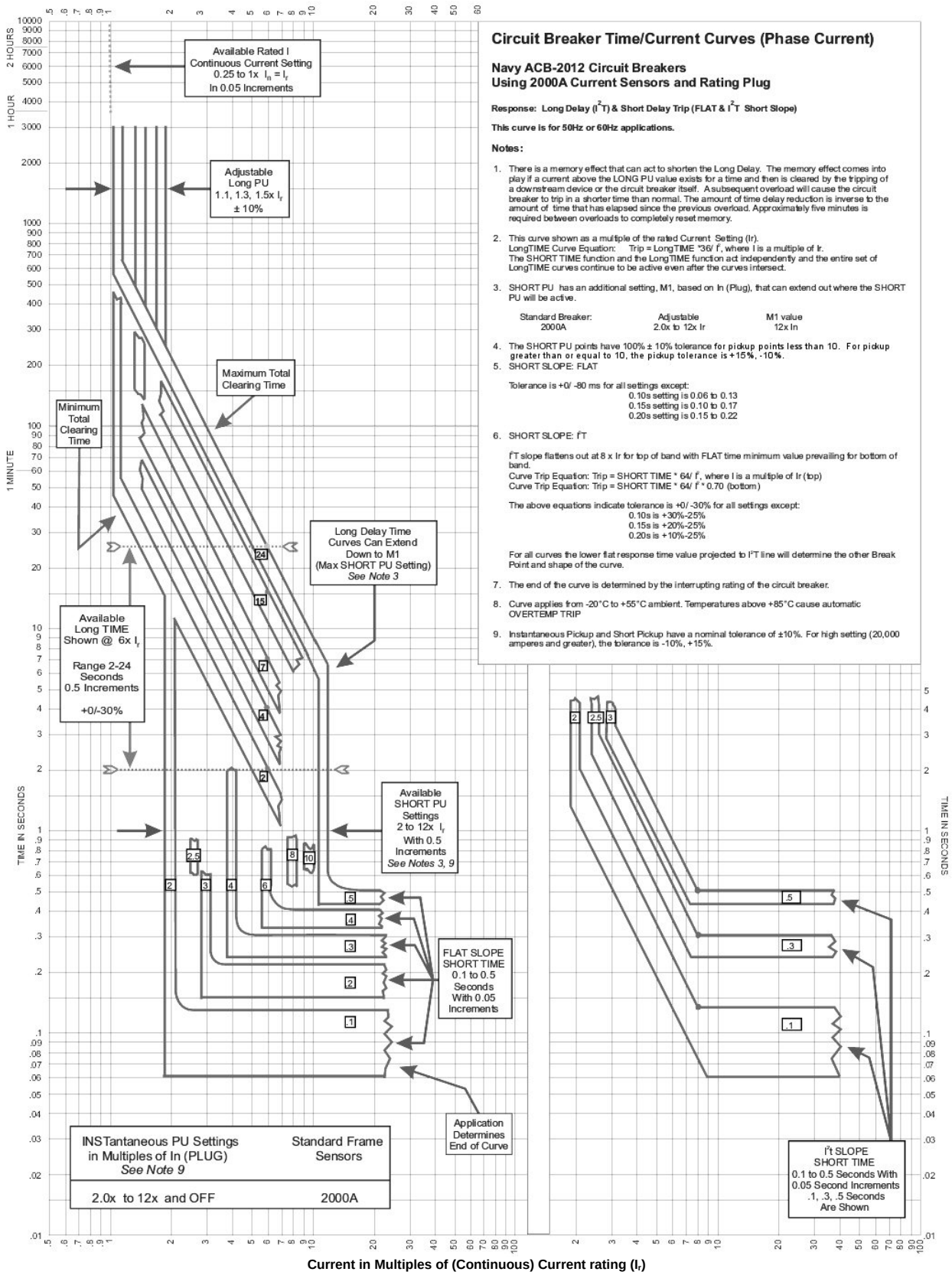


FIGURE 12. Time-current characteristic curve for circuit breaker type ACB-2012.

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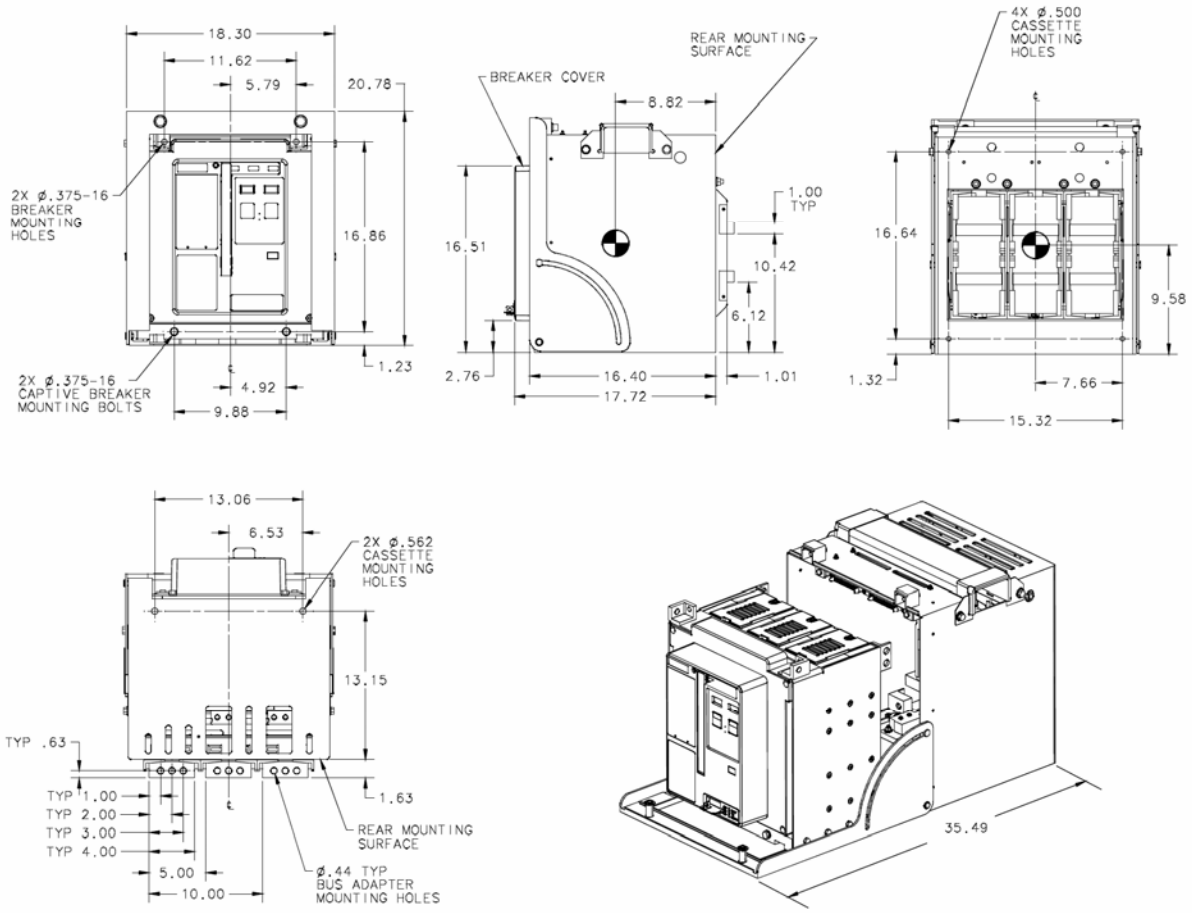


FIGURE 13. Outline and mounting dimensions for circuit breaker type ACB-2020.

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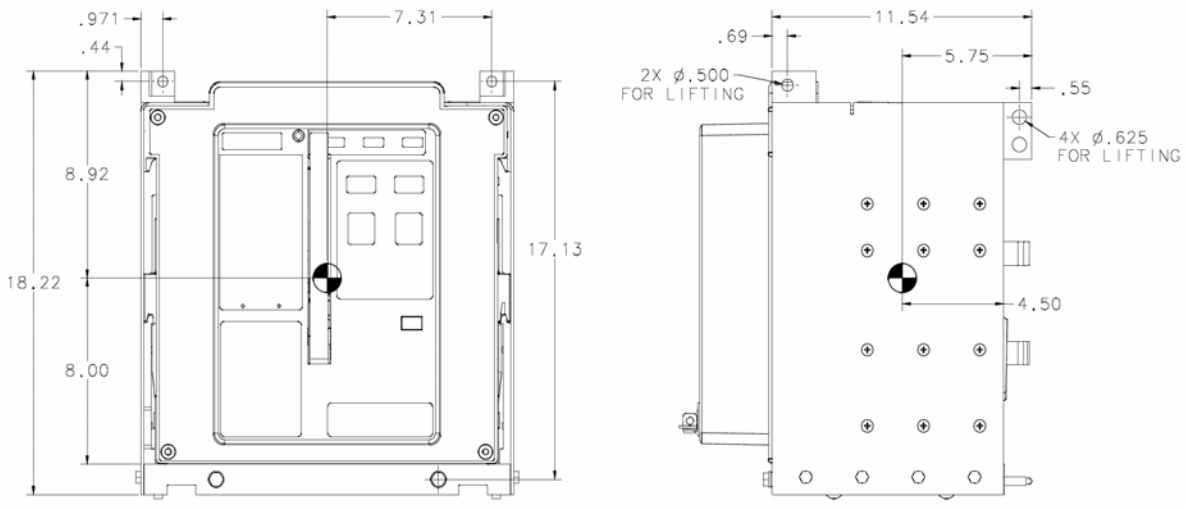


FIGURE 14. Outline and lift point dimensions for circuit breaker type ACB-2020 without cradle.

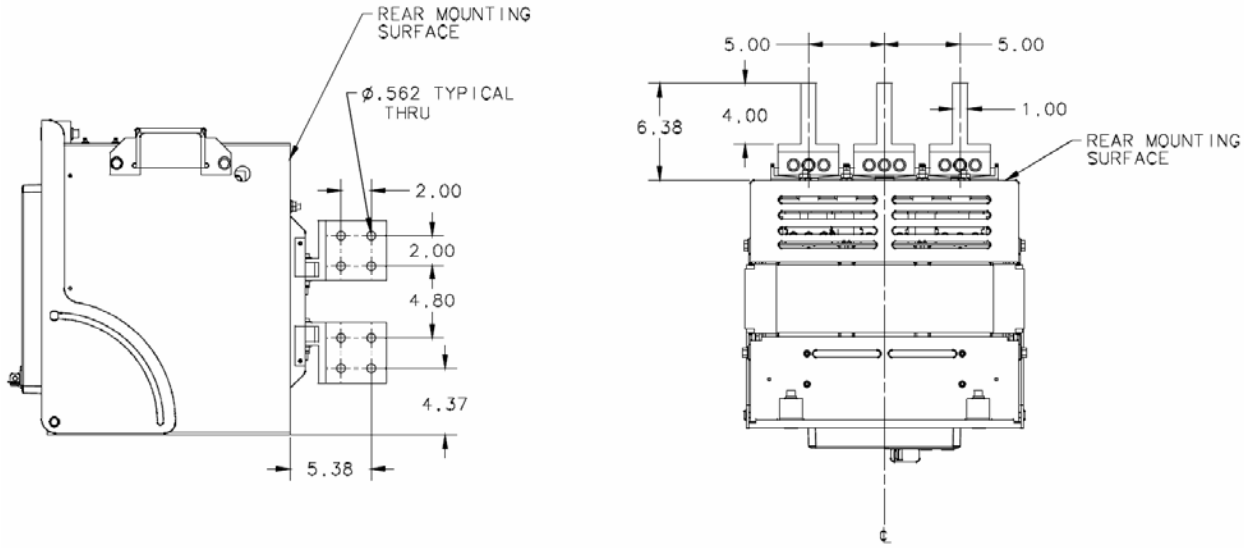
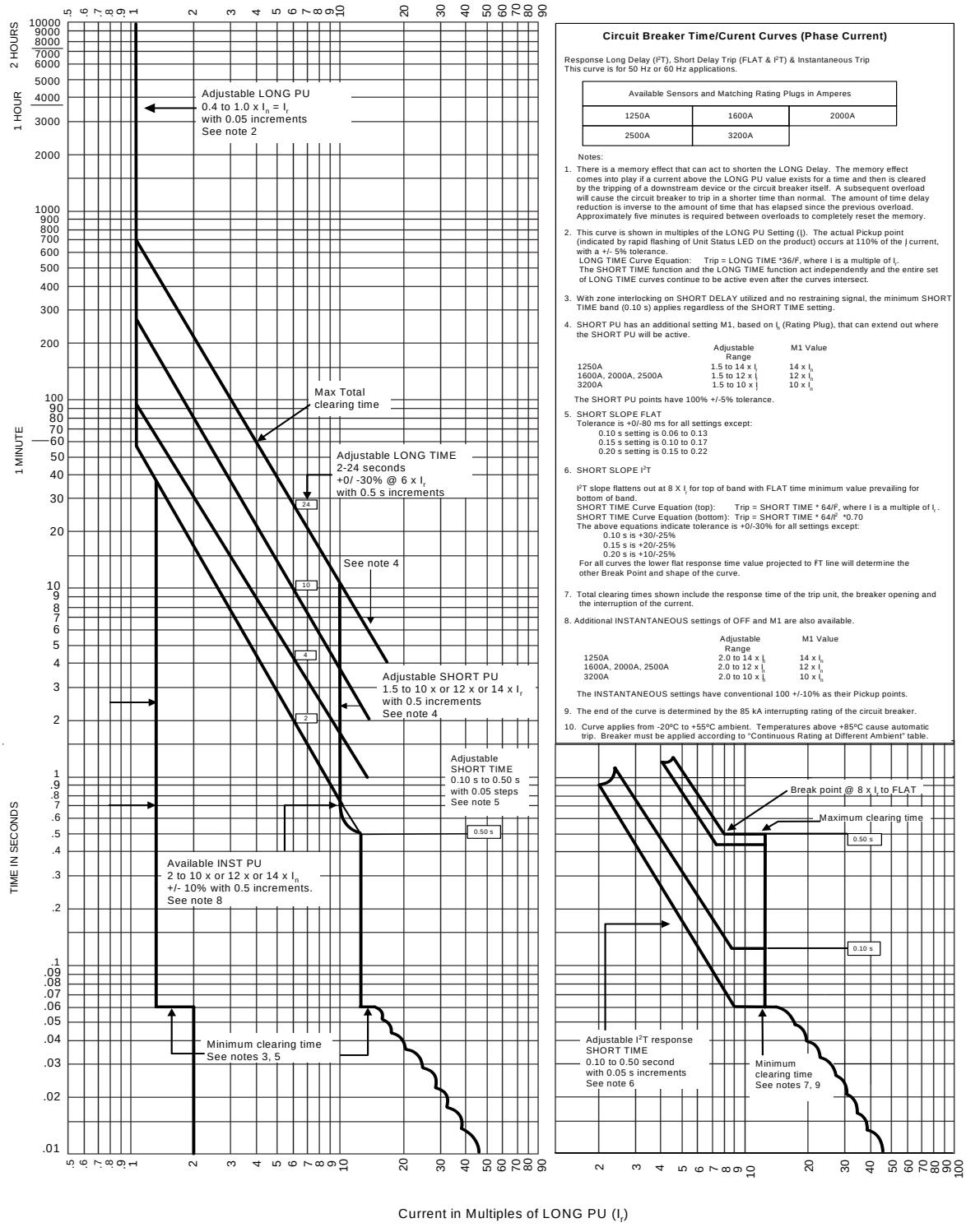


FIGURE 15. Connection dimensions for circuit breaker type ACB-2020 with optional vertical bus stab adapters.

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Circuit Breaker Time/Curent Curves (Phase Current)

Response Long Delay (I^2T), Short Delay Trip (FLAT & I^2T) & Instantaneous Trip
This curve is for 50 Hz or 60 Hz applications.

Available Sensors and Matching Rating Plugs in Amperes		
1250A	1600A	2000A
2500A	3200A	

Notes:

1. There is a memory effect that can act to shorten the LONG Delay. The memory effect comes into play if a current above the LONG PU value exists for a time and then is cleared by the tripping of a downstream device or the circuit breaker itself. A subsequent overload will cause the circuit breaker to trip in a shorter time than normal. The amount of time delay reduction is inverse to the amount of time that has elapsed since the previous overload. Approximately five minutes is required between overloads to completely reset the memory.
2. This curve is shown in multiples of the LONG PU Setting (I_r). The actual Pickup point (indicated by rapid flashing of Unit Status LED on the product) occurs at 110% of the I_r current, with a +/- 5% tolerance.
   LONG TIME Curve Equation: Trip = LONG TIME *36/I^2, where I is a multiple of I_r.
   The SHORT TIME function and the LONG TIME function act independently and the entire set of LONG TIME curves continue to be active even after the curves intersect.
3. With zone interlocking on SHORT DELAY utilized and no restraining signal, the minimum SHORT TIME band (0.10 s) applies regardless of the SHORT TIME setting.
4. SHORT PU has an additional setting M1, based on I_n (Rating Plug), that can extend out where the SHORT PU will be active.

	Adjustable Range	M1 Value
1250A	1.5 to 14 x I_r	14 x I_n
1600A, 2000A, 2500A	1.5 to 12 x I_r	12 x I_n
3200A	1.5 to 10 x I_r	10 x I_n

   The SHORT PU points have 100% +/-5% tolerance.
5. SHORT SLOPE FLAT
   Tolerance is +0/-80 ms for all settings except:
   0.10 s setting is 0.06 to 0.13
   0.15 s setting is 0.10 to 0.17
   0.20 s setting is 0.15 to 0.22
6. SHORT SLOPE I^2T

   I^2T slope flattens out at 8 X I_r for top of band with FLAT time minimum value prevailing for bottom of band.
   SHORT TIME Curve Equation (top): Trip = SHORT TIME * 64/I^2, where I is a multiple of I_r.
   SHORT TIME Curve Equation (bottom): Trip = SHORT TIME * 64/I^2 *0.70
   The above equations indicate tolerance is +0/-30% for all settings except:
   0.10 s is +30/-25%
   0.15 s is +20/-25%
   0.20 s is +10/-25%
   For all curves the lower flat response time value projected to FT line will determine the other Break Point and shape of the curve.
7. Total clearing times shown include the response time of the trip unit, the breaker opening and the interruption of the current.
8. Additional INSTANTANEOUS settings of OFF and M1 are also available.

	Adjustable Range	M1 Value
1250A	2.0 to 14 x I_n	14 x I_n
1600A, 2000A, 2500A	2.0 to 12 x I_n	12 x I_n
3200A	2.0 to 10 x I_n	10 x I_n

   The INSTANTANEOUS settings have conventional 100 +/-10% as their Pickup points.
9. The end of the curve is determined by the 85 kA interrupting rating of the circuit breaker.
10. Curve applies from -20ºC to +55ºC ambient. Temperatures above +85ºC cause automatic trip. Breaker must be applied according to "Continuous Rating at Different Ambient" table.

FIGURE 16. Time-current characteristic curve for circuit breaker type ACB-2020.

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Preparing Activity:
Navy – SH
(Project 5925-N184-003)

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